

Speakers



Mike Wishart, CEO,
e-Fabless



Dr Naveed Sherwani, CEO,
Rapid-Silicon



Dr Satya Gupta, President,
VLSI Society of India



Subhasish Mitra, Professor
at Stanford, the USA



Vivek Khaneja, Executive
Director, C-DAC

India first, world second

It is no surprise that the golden period for Indian electronics and semiconductor industry has started to take shape. Sherwani notes, "What we now need is a group of people, Indians who want to make India great. India has a very large number of people with extensive experience in various aspects of chip making. How many nations have that?"

Fostering this ripe talent and providing them avenues to come and build the semiconductor industry is essentially among the best ways to develop the same. Especially since most of these talents have worked closely with companies based in Silicon Valley, they have the knowledge and expertise to emulate the same in India. The product ecosystem in India is already well-developed, what it needs is a little fire to make it shine bright enough for the world to see.

The chip industry is a cutthroat one, and India being in its nascent stage to develop its own niche, needs to be wary of diving headlong

into the competition. While the ambition is understandable and lauded, common sense must still prevail if it wishes to make a mark. For that, it needs to turn its head and start looking inward rather than outward. "I think we must start with making simple chips, chips which are relevant to India," says Sherwani.

Microcontrollers, IoT chips, and LED PCBAs are some aspects that India can aim for, simply because their demand in local terms is extremely high, making it easily scalable while also giving the still-growing market a big indigenous push. Gupta suggests taking a thread out of the big-little approach that many MNCs have followed. This would allow Indian companies the option to choose indigenously produced chips to add to their domestically produced products, which would further increase their value addition.

"For instance, India today makes 700 (types of) LED bulbs domestically for domestic consumption. A small tensor chip goes into every small volt LED. It's a simple chip but the volume is humongous. And

since the product is also made by Indian companies, they can decide what to use. This could be the little part of the big-little. The big part would automatically proceed once we conquer our local ground," says Gupta.

Additionally, owing to the high investment that chip making requires, not every idea gets elevated. This also means that a lot of great ideas with little marketability, at least initially, have had to go down the drain only because someone decided it would not be worth spending on.

"I think we need to become more open and allow for more people to contribute their ideas in order to see what kind of stars can emerge. It is important to remove the ability of third parties to say 'no' to a project as much as we can," Wishart adds.

Grabbing the seat

The government's strong foot forward with regards to the semiconductor space was a much-needed move and, although late, it is expected to herald the golden era of the electronics industry of the country. What needs to be done is to adopt a problem-centric approach to think of an end-to-end solution, which can be scaled up from a local problem into a global problem. This is where the academia needs to play its part.

"The real question is how we can get real good ideas. The moment we are able to bring compelling ideas, people will be forced to fund them," reiterates Mitra.

While sovereignty is the ultimate goal, India needs to focus on getting a share of the pie, no matter how small it is. In the end, India needs to do enough to sit on the table with other countries, because if we control even 3% of the market, we can sit there. If we are zero, we will not even be invited. **EFY**

The author, Siddha Dhar, is a technology enthusiast at EFY